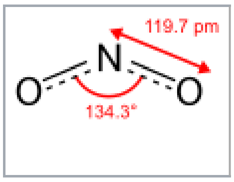
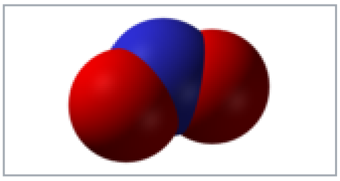
Nitrogen dioxide

Nitrogen dioxide is a chemical compound with the formula NO_2. It is one of several nitrogen oxides. NO_2 is an intermediate in the industrial synthesis of nitric acid, millions of tons of which are produced each year for use primarily in the production of fertilizers. At higher temperatures it is a reddish-brown gas. It can be fatal if inhaled in large quantities.[8] Nitrogen dioxide is a paramagnetic, bent molecule with C_{2v} point group symmetry.

It is included in the NO_x family of atmospheric pollutants.

Nitrogen dioxide





NO_2 converts to the colorless dinitrogen tetroxide (N_2O_4) at low temperatures and reverts to NO_2 at higher temperatures.

Names	
IUPAC name	Nitrogen dioxide
Other names	Nitrogen(IV) oxide,[1] deutoxide of nitrogen
Identifiers	
CAS Number	10102-44-0 (https://commonchemistry.cas.org/detail?cas_rn=10102-44-0) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=N%28%3DO%29%5BO%5D) Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=%5BN%2B%5D%28%3DO%29%5BO-%5D)

Properties

Nitrogen dioxide is a reddish-brown gas with a pungent, acrid odor above 21.2 °C (70.2 °F; 294.3 K) and becomes a yellowish-brown liquid below 21.2 °C (70.2 °F; 294.3 K). It forms an equilibrium with its dimer, dinitrogen tetroxide (N_2O_4), and converts almost entirely to N_2O_4below −11.2 °C (11.8 °F; 261.9 K).[6]

The bond length between the nitrogen atom and the oxygen atom is 119.7 pm. This bond length is consistent with a bond order between one and two.

Unlike ozone, O_3, the ground electronic state of nitrogen dioxide is a doublet state, since nitrogen has one unpaired electron,[9] which decreases the alpha effect compared with nitrite and creates a weak bonding interaction with the oxygen lone pairs. The lone electron in NO_2 also means that this compound is a free radical, so the formula for nitrogen dioxide is often written as $^{\bullet}NO_2$.

The reddish-brown color is a consequence of preferential absorption of light in the blue region of the spectrum (400 – 500 nm), although the absorption extends throughout the visible (at shorter wavelengths) and into the infrared (at longer wavelengths). Absorption of light at wavelengths shorter than about 400 nm results in photolysis (to form NO + O, atomic oxygen); in the atmosphere the addition of the oxygen atom so formed to O_2 results in ozone.

Preparation

Nitrogen dioxide typically arises via the oxidation of nitric oxide by oxygen in air (e.g. as result of corona discharge):[10]

$$NO + \tfrac{1}{2} O_2 \rightarrow NO_2$$

Nitrogen dioxide is formed in most combustion processes using air as the oxidant. At elevated temperatures nitrogen combines with oxygen to form nitrogen dioxide:

$$\tfrac{1}{2} N_2 + O_2 \rightarrow NO_2$$

In the laboratory, NO_2 can be prepared in a two-step procedure where dehydration of nitric acid produces dinitrogen pentoxide:

$$2\ HNO_3 \rightarrow N_2O_5 + H_2O$$
$$6\ HNO_3 + \tfrac{1}{2} P_4O_{10} \rightarrow 3\ N_2O_5 + 2\ H_3PO_4$$

which subsequently undergoes thermal decomposition:

$$N_2O_5 \rightarrow 2\ NO_2 + \tfrac{1}{2} O_2$$

The thermal decomposition of some metal nitrates also generates NO_2:

$$Pb(NO_3)_2 \rightarrow PbO + 2\ NO_2 + \tfrac{1}{2} O_2$$

NO_2 is generated by the reduction of concentrated nitric acid with a metal (such as copper):

$$4\ HNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2\ NO_2 + 2\ H_2O$$

Formation from decomposition of nitric acid

Nitric acid decomposes slowly to nitrogen dioxide by the overall reaction:

$$2\ HNO_3 \rightarrow 2\ NO_2 + H_2O + \tfrac{1}{2} O_2$$

ChEBI	CHEBI:33101 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=33101) ✓
ChemSpider	2297499 (https://www.chemspider.com/Chemical-Structure.2297499.html) ✓
ECHA InfoCard	100.030.234 (https://echa.europa.eu/substance-information/-/substanceinfo/100.030.234)
EC Number	233-272-6
Gmelin Reference	976
PubChem CID	3032552 (https://pubchem.ncbi.nlm.nih.gov/compound/3032552)
RTECS number	QW9800000
UNII	S7G510RUBH (https://precision.fda.gov/uniisearch/srs/unii/S7G510RUBH) ✗
UN number	1067
CompTox Dashboard (EPA)	DTXSID7020974 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID7020974)
InChI	
InChI=1S/NO2/c2-1-3 ✓ Key: JCXJVPUVTGWSNB-UHFFFAOYSA-N ✓	
InChI=1/NO2/c2-1-3 Key: JCXJVPUVTGWSNB-UHFFFAOYAA	
SMILES	
N(=O)[O]	
[N+](=O)[O-]	
Properties	
Chemical formula	$NO_2^{\bullet}$
Molar mass	46.006 g/mol[2]
Appearance	Brown gas[2]
Odor	Chlorine-like
Density	1.880 g/L[2]

The nitrogen dioxide so formed confers the characteristic yellow color often exhibited by this acid.

Selected reactions

Thermal properties

NO_2 exists in equilibrium with the colourless gas dinitrogen tetroxide (N_2O_4):

$$2\ NO_2 \rightleftharpoons N_2O_4$$

The equilibrium is characterized by $\Delta H = -57.23$ kJ/mol, which is exothermic. NO_2 is favored at higher temperatures, while at lower temperatures, N_2O_4 predominates. N_2O_4 can be obtained as a white solid with melting point −11.2 °C.[10] NO_2 is paramagnetic due to its unpaired electron, while N_2O_4 is diamagnetic.

At 150 °C, NO_2 decomposes with release of oxygen via an endothermic process (ΔH = 14 kJ/mol):

$$NO_2 \rightarrow NO + \tfrac{1}{2}\ O_2$$

As an oxidizer

As suggested by the weakness of the N–O bond, NO_2 is a good oxidizer. Consequently, it will combust, sometimes explosively, in the presence of hydrocarbons.

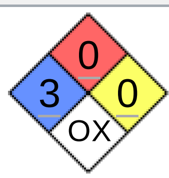
Hydrolysis

NO_2 reacts with water to give nitric acid and nitrous acid:

$$2\ NO_2 + H_2O \rightarrow HNO_2 + HNO_3$$

This reaction is one of the steps in the Ostwald process for the industrial production of nitric acid from ammonia.[11] This reaction is negligibly slow at low concentrations of NO_2 characteristic of the ambient atmosphere, although it does proceed upon NO_2 uptake to surfaces. Such surface reaction is thought to produce gaseous HNO_2 (often written as HONO) in outdoor and indoor environments.[12]

Conversion to nitrates

Melting point	−9.3 °C (15.3 °F; 263.8 K)[2]
Boiling point	21.15 °C (70.07 °F; 294.30 K)[2]
Solubility in water	Hydrolyses
Solubility	Soluble in CCl_4, nitric acid,[3] chloroform
Vapor pressure	98.80 kPa (at 20 °C)
Magnetic susceptibility (χ)	+150.0·10^{-6} cm^3/mol[4]
Refractive index (n_D)	1.449 (at 20 °C)
Structure	
Point group	C_{2v}
Molecular shape	Bent
Thermochemistry[5]	
Heat capacity (*C*)	37.2 J/(mol·K)
Std molar entropy ($S^{\ominus}_{298}$)	240.1 J/(mol·K)
Std enthalpy of formation ($\Delta_f H^{\ominus}_{298}$)	+33.2 kJ/mol
Hazards	
Occupational safety and health (OHS/OSH):	
Main hazards	Poison, oxidizer
GHS labelling:	
Pictograms	
Signal word	**Danger**
Hazard statements	H270, H314, H330
Precautionary statements	P220, P260, P280, P284, P305+P351+P338, P310
NFPA 704 (fire diamond)	
Lethal dose or concentration (LD, LC):	

NO_2 is used to generate anhydrous metal nitrates from the oxides:[10]

$$MO + 3\ NO_2 \rightarrow M(NO_3)_2 + NO$$

Alkyl and metal iodides give the corresponding nitrates:[9]

$$TiI_4 + 8\ NO_2 \rightarrow Ti(NO_3)_4 + 4\ NO + 2\ I_2$$

Ecology

NO_2 is introduced into the environment by natural causes, including entry from the stratosphere, bacterial respiration, volcanos, and lightning. These sources make NO_2 a trace gas in the atmosphere of Earth, where it plays a role in absorbing sunlight and regulating the chemistry of the troposphere, especially in determining ozone concentrations.[13]

Uses

NO_2 is used as an intermediate in the manufacturing of nitric acid, as a nitrating agent in the manufacturing of chemical explosives, as a polymerization inhibitor for acrylates, as a flour bleaching agent,[14]:223 and as a room temperature sterilization agent.[15] It is also used as an oxidizer in rocket fuel, for example in red fuming nitric acid; it was used in the Titan rockets, to launch Project Gemini, in the maneuvering thrusters of the Space Shuttle, and in uncrewed space probes sent to various planets.[16]

LC_{50} (median concentration)	30 ppm (guinea pig, 1 h) 315 ppm (rabbit, 15 min) 68 ppm (rat, 4 h) 138 ppm (rat, 30 min) 1000 ppm (mouse, 10 min)[7]
LC_{Lo} (lowest published)	64 ppm (dog, 8 h) 64 ppm (monkey, 8 h)[7]
NIOSH (US health exposure limits):	
PEL (Permissible)	C 5 ppm (9 mg/m^3)[6]
REL (Recommended)	ST 1 ppm (1.8 mg/m^3)[6]
IDLH (Immediate danger)	13 ppm[6]
Safety data sheet (SDS)	ICSC 0930 (http://www.inchem.org/documents/icsc/icsc/eics0930.htm)
Related compounds	
Related nitrogen oxides	Dinitrogen pentoxide Dinitrogen tetroxide Dinitrogen trioxide Nitric oxide Nitrous oxide
Related compounds	Chlorine dioxide Carbon dioxide
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).	
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Infobox references	

Human-caused sources and exposure

A "fox tail" over Nizhniy Tagil Iron and Steel Works

For the general public, the most prominent sources of NO_2 are internal combustion engines, as combustion temperatures are high enough to thermally combine some of the nitrogen and oxygen in the air to form NO_2.[8] Outdoors, NO_2 can be a result of traffic from motor vehicles.[17]

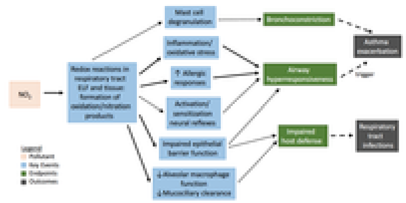
Indoors, exposure arises from cigarette smoke,[18] and butane and kerosene heaters and stoves.[19]

Workers in industries where NO_2 is used are also exposed and are at risk for occupational lung diseases, and NIOSH has set exposure limits and safety standards.[6] Workers in high voltage areas especially those with spark or plasma creation are at risk. Agricultural workers can be exposed to NO_2 arising from grain decomposing in silos; chronic exposure can lead to lung damage in a condition called "silo-filler's disease".[20][21]

Toxicity

NO_2 diffuses into the epithelial lining fluid (ELF) of the respiratory epithelium and dissolves. There, it chemically reacts with antioxidant and lipid molecules in the ELF. The health effects of NO_2 are caused by the reaction products or their metabolites, which are reactive nitrogen species and reactive oxygen species that can drive bronchoconstriction, inflammation, reduced immune response, and may have effects on the heart.[22]

Acute harm due to NO_2 exposure is rare. 100–200 ppm can cause mild irritation of the nose and throat, 250–500 ppm can cause edema, leading to bronchitis or pneumonia, and levels above 1000 ppm can cause death due to asphyxiation from fluid in the lungs. There are often no symptoms at the time of exposure other than transient cough, fatigue or nausea, but over hours inflammation in the lungs causes edema.[23][24]



Pathways indicated by a dotted line are those for which evidence is limited to findings from experimental animal studies, while evidence from controlled human exposure studies is available for pathways indicated by a solid line. Dashed lines indicate proposed links to the outcomes of asthma exacerbation and respiratory tract infections. Key events are subclinical effects, endpoints are effects that are generally measured in the clinic, and outcomes are health effects at the organism level. NO_2 = nitrogen dioxide; ELF = epithelial lining fluid.[22]:4–62

For skin or eye exposure, the affected area is flushed with saline. For inhalation, oxygen is administered, bronchodilators may be administered, and if there are signs of methemoglobinemia, a condition that arises when nitrogen-based compounds affect the hemoglobin in red blood cells, methylene blue may be administered.[25][26]

It is classified as an extremely hazardous substance in the United States as defined in Section 302 of the U.S. Emergency Planning and Community Right-to-Know Act (42 U.S.C. 11002), and it is subject to strict reporting requirements by facilities which produce, store, or use it in significant quantities.[27]

Health effects of NO_2 exposure

Even small day-to-day variations in NO_2 can cause changes in lung function.[28] Chronic exposure to NO_2 can cause respiratory effects including airway inflammation in healthy people and increased respiratory symptoms in people with asthma.

The effects of toxicity on health have been examined using questionnaires and in-person interviews in an effort to understand the relationship between NO_2 and asthma. The influence of indoor air pollutants on health is important because the majority of people in the world spend more than 80% of their time indoors.[29] The amount of time spent indoors depends upon on several factors including geographical region, job activities, and gender among other variables. Additionally, because home insulation is improving, this can result in greater retention of indoor air pollutants, such as NO_2.[29] With respect to geographic region, the prevalence of asthma has ranged from 2 to 20% with no clear indication as to what's driving the difference.[29] This may be a result of the "hygiene hypothesis" or "western lifestyle" that captures the notions of homes that are well insulated and with fewer inhabitants.[29] Another study examined the relationship between nitrogen exposure in the home and respiratory symptoms and found a statistically significant odds ratio of 2.23 (95% CI: 1.06, 4.72) among those with a medical diagnosis of asthma and gas stove exposure.[30]

Nitrogen dioxide diffusion tube for air quality monitoring. Positioned in the City of London.

A major source of indoor exposure to NO_2 is the use of gas stoves for cooking or heating in homes. According to the 2000 census, over half of US households use gas stoves[31] and indoor exposure levels of NO_2 are, on average, at least three times higher in homes with gas stoves compared to electric stoves with the highest levels being in multifamily homes. Exposure to NO_2 is especially harmful for children with asthma. Research has shown that children with asthma who live in homes with gas stoves have greater risk of respiratory symptoms such as wheezing, cough and chest tightness.[30][32] Additionally, gas stove use was associated with reduced lung function in girls with asthma, although this association was not found in boys.[33] Using ventilation when operating gas stoves may reduce the risk of respiratory symptoms in children with asthma.

In a cohort study with inner-city minority African American Baltimore children to determine if there was a relationship between NO_2 and asthma for children aged 2 to 6 years old, with an existing medical diagnosis of asthma, and one asthma related visit, families of lower socioeconomic status were more likely to have gas stoves in their homes. The study concluded that higher levels of NO_2 within a home were linked to a greater level of respiratory symptoms among the study population. This further exemplifies that NO_2 toxicity is dangerous for children.[34]

Environmental effects

Interaction of NO_2 and other NO_x with water, oxygen and other chemicals in the atmosphere can form acid rain which harms sensitive ecosystems such as lakes and forests.[35] Elevated levels of NO_2 can also harm vegetation, decreasing growth, and reduce crop yields.[36]

See also

- Dinitrogen tetroxide
- Nitric oxide (NO) – pollutant that is short lived because it converts to NO_2 in the presence of ozone
- Nitrite
- Nitrous oxide (N_2O) – "laughing gas", a linear molecule, isoelectronic with CO_2 but with a nonsymmetric arrangement of atoms (NNO)
- Nitryl

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External links

- International Chemical Safety Card 0930 (http://www.inchem.org/documents/icsc/icsc/eics0930.htm)
- National Pollutant Inventory – Oxides of nitrogen fact sheet (http://www.npi.gov.au/resource/oxides-nitrogen-0)
- NIOSH Pocket Guide to Chemical Hazards (https://www.cdc.gov/niosh/npg/npgd0454.html)
- WHO-Europe reports: Health Aspects of Air Pollution (2003) (http://www.euro.who.int/document/e79097.pdf) (PDF) and "Answer to follow-up questions from CAFE (2004) (https://web.archive.org/web/20050909162153/http://www.euro.who.int/document/E82790.pdf) (PDF)
- Nitrogen Dioxide Air Pollution (http://www.greenfacts.org/air-pollution/nitrogen-dioxide-no2/index.htm)
- Current global map of nitrogen dioxide distribution (https://earth.nullschool.net/#current/chem/surface/level/overlay=no2/winkel3)

- A review of the acute and long term impacts of exposure to nitrogen dioxide in the United Kingdom (https://web.archive.org/web/20160415110253/http://www.iom-world.org/pubs/IOM_TM0403.pdf) IOM Research Report TM/04/03
